

Vishal Kumar Mathuri

BLOCKCHAIN ENGINEER | SOLIDITY / EVM | RUST / SOLANA

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SUMMARY

Blockchain Engineer with hands-on professional experience building **Solidity smart contracts, EVM protocols, DeFi systems, tokenized applications**, and full-stack Web3 products — including **50+ smart contracts** and **10+ full-stack blockchain applications** shipped to date. Strong focus on **secure and testable smart-contract engineering** using Solidity, Foundry, Hardhat, OpenZeppelin, and Ethers.js, with full-stack experience across React.js, Node.js, and REST APIs.

Expanding into **Rust/Solana and cross-chain protocol engineering**, with a portfolio centered on **production-oriented protocol design** rather than a collection of tutorial projects.

TECHNICAL SKILLS

Smart Contracts: Solidity, ERC-20, ERC-721, access control, staking, governance, vesting, custom errors, events

EVM / Blockchain: Ethereum, EVM, Polygon, Ethereum Sepolia, gas optimization, contract verification, IPFS

Security / Testing: OpenZeppelin, ReentrancyGuard, AccessControl, Ownable, Pausable, unit testing, fuzz testing, invariant testing

Web3 / Backend: Ethers.js, Web3.js, React.js, Node.js, Express.js, REST APIs, MetaMask

Rust / Solana: Rust, Solana, Anchor, PDAs, TypeScript — actively developing production-oriented Solana projects

Tools: Foundry, Hardhat, Remix, Git, GitHub, VS Code, Vercel, Render, IPFS/Pinata

PROFESSIONAL EXPERIENCE

Blockchain Developer Freelance	Jul 2025 - Present
- Developed and deployed 50+ Solidity smart contracts implementing ERC-20, ERC-721, staking, governance, and marketplace functionality across NFT, fundraising, identity, and access-control applications. - Built 10+ full-stack Web3 applications integrating smart contracts with React.js frontends, Node.js backend services, REST APIs, wallets, and EVM networks. - Applied OpenZeppelin security primitives and secure design patterns including reentrancy protection and role-based access control, reducing security risk across deployed contracts. - Worked across contract implementation, testing, deployment, frontend integration, debugging, and production issue resolution.	
Senior Content Developer Think & Learn Pvt Ltd	Jul 2019 - Mar 2023 Bengaluru, IN
- Managed scheduling operations for 200+ student batches across multiple grade levels with 100% operational accuracy and enhanced workflow efficiency. - Improved an automated concept-tagging workflow to 95% classification accuracy through systematic validation and quality checks, boosting system efficiency and reducing errors. - Crafted and executed structured Learning Journeys for academic programs utilizing digital learning platforms, resulting in a 96% success rate. - Designed and curated interactive educational content, mock tests, and learning modules for Byju's TV and BHLp programs, increasing user satisfaction and engagement. - Resolved student support and academic operations requests using Salesforce with 98% resolution accuracy within 24 hours, ensuring uninterrupted course delivery. - Collaborated with academic operations, HR, and support teams while maintaining strong cross-team communication and coordination.	

FLAGSHIP BLOCKCHAIN PROJECTS

1. Production-Grade EVM Lending Protocol

Tech Stack: Solidity • Foundry • Chainlink • OpenZeppelin

Links: [GitHub](#)

- Engineered a modular, overcollateralized EVM lending protocol supporting multi-asset supply, withdrawal, borrowing, repayment, interest accrual, and liquidation, with responsibilities separated across **LendingPool, ReserveManager, RiskManager**, and configurable interest-rate modules.
- Implemented **index-based scaled accounting and risk management**, including utilization-driven borrow/supply rates, collateral-factor borrowing limits, health-factor solvency checks, liquidation thresholds/bonuses, multi-decimal asset normalization, and rounding-aware debt/collateral accounting.
- Built a **Chainlink-based oracle layer** with defensive price validation, including stale-price protection, incomplete/stale round detection, positive-price validation, future-timestamp checks, configurable feed heartbeats, and an oracle abstraction enabling deterministic mock testing.
- Validated the protocol with an extensive **Foundry test suite spanning unit, integration, fuzz, and stateful invariant testing**, plus failure-path and security suites; deployed and source-verified the contracts on Ethereum Sepolia and executed an end-to-end supply → borrow → repay → withdraw flow.

2. NFT Marketplace

Tech Stack: Solidity • ERC-721 • OpenZeppelin • React.js • Ethers.js • IPFS/Pinata

Links: [GitHub](#) • [Live \(Vercel\)](#) • [API \(Render\)](#)

- Engineered a full-stack NFT marketplace supporting ERC-721 minting, listing, purchasing, and ownership transfers through Solidity smart contracts, with a React.js interface for end-to-end marketplace interactions.
- Integrated **Ethers.js and MetaMask** for wallet-based Web3 workflows, enabling users to connect accounts, sign transactions, mint NFTs, list assets, and execute purchases directly against deployed smart contracts.
- Implemented decentralized NFT metadata storage using **IPFS/Pinata**, linking token metadata and digital assets to ERC-721 tokens while using OpenZeppelin contracts for standardized NFT functionality and ownership controls.

- Delivered a publicly accessible full-stack application, deploying the React frontend on Vercel and backend services on Render to provide a live, testable Web3 application beyond standalone smart contracts.

3. Solana Crowdfunding Protocol

Tech Stack: Rust • Solana • Anchor • PDAs • TypeScript • Wallet Adapter

Status: In Development Links: [GitHub](#) • [Live \(Vercel\)](#)

- Building a Solana crowdfunding protocol in Rust and Anchor for campaign creation, SOL contributions, funding-goal tracking, deadline enforcement, creator withdrawals, and contributor refunds.
- Designing **PDA-based campaign and protocol state**, using deterministic account derivation to associate campaign data and protocol-controlled accounts with campaign creators without relying on externally generated private keys.
- Implementing Anchor account constraints and authorization rules to validate campaign ownership, signer permissions, contribution state, withdrawal eligibility, and refund conditions at the program level.
- Developing the TypeScript/Wallet Adapter integration layer for wallet-connected campaign interactions while structuring tests around successful funding, failed campaigns, unauthorized withdrawals, deadline boundaries, and refund behavior.

4. LayerZero Cross-Chain Protocol (Ethereum ↔ Solana)

Tech Stack: Solidity • Rust • LayerZero V2 • Ethereum • Solana • Anchor • Foundry • React • Ethers.js • Solana Web3.js

Links: [GitHub](#) • [Live Demo](#)

- Engineered a bidirectional **LayerZero V2 interoperability protocol** connecting Ethereum Sepolia and Solana Devnet, supporting cross-chain application messaging and **OFT (Omnichain Fungible Token) transfers** between EVM and Solana environments.
- Built a unified **MetaMask + Phantom wallet frontend** for cross-chain interactions, including a Phantom ALT (Address Lookup Table) compatibility workaround to handle Solana transaction size constraints.
- Implemented **LayerZero fee quoting and delivery tracking** alongside Solana transaction simulation and replay protection, ensuring reliable and verifiable message delivery across chains.
- Delivered peer validation, message verification, and versioned transaction handling across Solidity and Rust/Anchor components, with end-to-end deployment and testing on Ethereum Sepolia and Solana Devnet.

INTERNSHIPS

Blockchain Developer EtherAuthority	Mar 2026 - May 2026 Remote
• Built a Telegram Mini App featuring multiple in-app mini-games, including Coinflip, Find the King (a dice-based game), and Wheel, using the Telegram Web App platform. • Designed and implemented game logic, betting flow, and result computation for each game mode. • Collaborated with the team on frontend/backend integration and testing to ship engaging, low-latency gameplay experiences.	
Blockchain Developer Leafny Web Pack	May 2025 - Jul 2025 Remote
• Assisted in building Web3 dApps integrating smart contracts with frontend and backend systems. • Worked with ERC-20 and ERC-721 token standards and participated in smart-contract security testing and reviews.	
Blockchain Developer Estatos	Jan 2025 - Feb 2025 Remote
• Developed and optimized smart contracts for blockchain operations. • Worked on tokenization and DeFi protocols, contributing to secure and efficient smart-contract deployments. • Collaborated with the tech team to enhance solutions while demonstrating problem-solving and a growth mindset.	
Solidity Developer Nexus Network	Jul 2024 - Dec 2024 Remote
• Developed bridge contracts for relayer-based cross-chain transactions. • Strengthened smart-contract security and optimization practices. • Actively participated in team collaborations and implemented feedback for continuous improvement.	

EDUCATION

Post Graduate Diploma in Blockchain IIIT Bangalore & upGrad	Mar 2023 - Mar 2024 Bengaluru, IN
• Completed advanced coursework covering JavaScript, Node.js, Ethereum smart-contract development, and Hyperledger Fabric. • Developed Ethereum-based smart-contract architectures for decentralized applications including DeFi systems. • Designed system architecture for the UPSTAC healthcare application, focusing on scalable and secure blockchain integration. • Explored blockchain-based KYC systems, Hyperledger Fabric network setup, and cross-chain frameworks.	
Dual Degree (B.Tech + M.Tech), Energy Science & Engineering Indian Institute of Technology Bombay (IIT Bombay)	Jul 2014 - Jul 2019 Mumbai, IN
• Secured 61%.	

ADDITIONAL INFORMATION

- **Languages:** Fluent in Hindi (Native) and English (Professional)
- **Interests:** Reading and Cycling